Product Requirements Document: Vijayarka Nanduri Personal Website

Document status: Build specification for a first complete release
Prepared: 10 November 2026
Product: Vijayarka Nanduri — personal brand website and online studio
Primary audience: collaborators, researchers, designers, developers, potential clients, readers, and curious visitors
Primary outcome: a beautiful, memorable, readable website that presents Vijayarka's identity and body of work as one coherent world, while letting visitors quickly choose whether to watch a short visual introduction or explore directly.

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1. Product brief

Build a polished, responsive personal website for Vijayarka Nanduri. It should feel like entering a cinematic creative studio: emotionally distinctive and full of small discoveries, but clear and easy to navigate. The site is both a personal brand homepage and an "online office" for distinct types of work: projects, research, creative work, biography, and contact.

The site is **not** a generic developer résumé, an unfiltered GitHub clone, a forced movie intro, or a collection of unrelated visual themes. It has one recognizable brand system, with different sections behaving like different rooms in one house. Each room may have its own accent and material, but shared typography, spacing, navigation, contrast, and motion must keep them connected.

The site offers two equal, understandable routes into the work:

1. **Explore:** use the site normally and go directly to projects, research, creative work, or contact.
2. **Watch:** play a short, optional, sound-off-by-default visual trailer introducing Vijayarka and his work.

The trailer is a feature inside the site, not a video replacement for the site. It must never block navigation or content. On a phone, the trailer may invite a landscape rotation for the 16:9 composition, but portrait browsing and skipping must remain available.

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2. Brand and experience principles

2.1 Brand idea

A house of living ideas: research, software, and creative work are different rooms in Vijayarka’ s world. Each has its own character while sharing a common visual DNA.

2.2 Experience principles

- **Cinematic, not obstructive.** Use framing, pace, image, typography, and restrained movement to create atmosphere. Never force a long intro, scroll-jack, hide navigation, or delay access to information.
- **Playful, not chaotic.** Keep the wit and curiosity of the current site, but use playful copy selectively. Do not make every heading a joke or every section a terminal interface.
- **Established, not generic.** The opening screen should immediately show his full name, a clear professional/creative identity, and meaningful paths into the site.
- **Distinct, not disconnected.** Visual variation is intentional and limited to section accents/materials. Every section retains common shell, typography, spacing, button language, and transitions.
- **Readable before decorative.** Strong contrast, meaningful headings, short paragraphs, visible links, responsive type, accessible focus, and keyboard support take priority over effects.
- **Curated, not exhaustive on the homepage.** Feature a small selection, and let the Projects area function as a browsable catalogue for the wider portfolio.
- **Honest about status.** Every project has a truthful status and working action. Do not claim peer review, efficacy, privacy, or availability beyond verified evidence.

2.3 Brand voice

Warm, intelligent, curious, lightly witty, plain-spoken. Confident but not grandiose. Use first person on About and creator statements. Avoid unsubstantiated superlatives such as “revolutionary,” “world-first,” or “proven neuroscience.” Use jokes as brief asides rather than replacing product explanations.

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3. Users and core journeys

3.1 Visitor groups

- A first-time visitor who wants to know who Vijayarka is within a few seconds.
- A project visitor who wants to try a web app or find an Android APK/source repository.
- A research visitor who wants a reliable paper summary, DOI, file, and limitations.
- A creative-work visitor who wants to browse books, games, audio, or writing.

- A professional visitor who wants to understand fit and get in touch.
- A returning visitor who wants to find a specific project or recent release quickly.

3.2 Core journeys and success conditions

First-time arrival: sees name + clear positioning + Explore and Play options; can proceed without waiting or dismissing a gate.

Watch intro: opens accessible 16:9 trailer; can pause, mute/unmute, skip, close, and follow a final CTA to projects, research, or contact. No autoplay with sound.

Find an app: enters Projects, filters by platform/category/status, opens a detail page, and reaches the live product/download/source without ambiguity.

Read research: enters Research, understands the subject and evidence limits from a short summary, then opens the DOI/preprint.

Contact: reaches the contact area from navigation and page endings, understands the available contact path, and can use it on mobile.

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4. Information architecture and routes

Use these human-readable, stable routes. Keep navigation labels and route names consistent. A route can be implemented as a page or a section with a stable anchor, but Projects, Research, and Creative need index pages; substantial individual work needs detail pages.

- `/` — Home
- `/projects/` — project catalogue
- `/projects/mawo/`
- `/projects/khata/`
- `/projects/metabasis/`
- `/projects/almara/`
- `/projects/examflow/`
- `/projects/centrum-os/`
- `/projects/tenet/`
- `/projects/designrx/`
- `/projects/draftbook/`
- `/projects/echo/`

- `/projects/ek-indian-bacha/`
- `/projects/logbook/`
- `/projects/writers-assistant/`
- `/projects/tailored-poems/`
- `/projects/sylph/` (optional, only if sufficiently documented and owner-selected)
- `/research/` — research index
- `/research/designing-alive-ui/`
- `/research/self-narrating-neural-networks/`
- `/research/emotion-aware-ai-writing-tools/`
- `/creative/` — books, games, stories, audio and other creative work
- `/about/` — biography, principles, timeline
- `/contact/` — contact and collaboration
- `/privacy/` — only if site uses forms, analytics, or other personal-data processing; describe actual behavior, not boilerplate
- `/404.html` (or host-equivalent) — branded not-found page with useful routes

Do not publish a page as if a private or unannounced product is available. A future feature may be held in an unpublished CMS/data entry until Vijayarka explicitly approves public launch.

Global navigation

Desktop: brand wordmark at left; **Projects, Research, Creative, About, Contact** as visible links; **Play intro** as a visually distinct but not overpowering action. The header may become translucent on scroll, but must retain readable contrast.

Mobile: wordmark, a compact menu button with accessible name, and a prominent but compact Play intro entry if space permits. Menu opens a readable overlay/drawer with large tap targets; focus is contained while open and restored on close. Always include visible close, route, and current-page indication.

Footer: short signature/tagline, navigation, selected social/research links, and a sensible copyright year generated dynamically. Include only approved public contact details.

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5. Visual system

5.1 Overall art direction

Create an **editorial cinematic studio**: the emotional clarity of a film title sequence, the useful structure of a thoughtfully designed product site, and a few tactile surprises. Use generous negative space, strong image crops, soft changes of light, varied composition, and staged but short reveals. The site should feel beautiful and intentional even when animations are disabled.

Do not use generic dashboard styling as the entire brand. Material UI is a component/toolkit input, not the identity. Liquid glass, neural brutalism, and papercraft are controlled materials used for particular content rooms and elements—not literal visual effects applied everywhere.

5.2 Shared color tokens

Use the following exact initial tokens. Treat these as a coherent family and implement them as CSS custom properties and corresponding component-theme tokens. Validate contrast and adjust only if the stated accessibility contrast is preserved.

Core brand

- `--ink-950: #171522` — primary dark text / deepest ink
- `--ink-800: #2B2938` — secondary dark surface
- `--paper-50: #FBF9F5` — main light page background
- `--paper-100: #F3EFE7` — secondary paper surface
- `--paper-200: #E7E0D5` — borders and paper depth
- `--white: #FFFFFF` — text on dark / clean contrast surface
- `--brand-violet: #7657F6` — primary brand accent and main interactive action
- `--brand-violet-dark: #5A3BD0` — accessible hover/text variation
- `--brand-cyan: #74D7E8` — signal/light accent for highlights; do not use for small text on white
- `--brand-lime: #C9E76B` — rare energetic highlight / status detail, never body text
- `--muted-ink: #666271` — body secondary text on light background; check actual contrast
- `--line: #DDD7CE` — light dividers
- `--focus: #1F72D2` — keyboard focus ring

Room accents

- Home / overall brand: violet `#7657F6` with cyan `#74D7E8` as a restrained signal.
- Projects / workshop: green `#53886D` with pale green surface `#EAF1E9`.
- Research / observatory: blue `#456C9A` with pale blue surface `#EAF0F6`.

- Creative / studio: amber `#B7793E` with pale amber surface `#F6EBDD`.
- About / personal room: plum `#80617F` with pale plum surface `#F1E9F0`.
- Contact / invitation: coral `#C96D63` with pale coral surface `#F7EAE7`.

Room accents may tint a section line, chip, graphic, subtle background wash, or CTA. Do not recolor the entire page or change the core text/background palette from room to room. Support the same light/dark brand continuity if a dark theme is implemented; dark theme is optional for first release and must not delay a strong light design.

5.3 Theme balance

The default experience should be the light paper theme because it supports long reading and lets project imagery breathe. Use ink/dark surfaces for selected trailer frames, project imagery, or a contained "night" area. Do not alternate every section between unrelated full-screen color blocks. Maintain a dominant neutral field with accents creating atmosphere.

5.4 Typography

Use locally hosted/self-hosted variable fonts where licensing permits, with system fallbacks. Avoid loading fonts from a third-party runtime if bundled files are available.

- **Display / major headings:** `Space Grotesk`, variable weights 400–700. Expressive, technical, but approachable. Use for hero name, large section titles, project titles, and short labels.
- **Body / reading:** `Manrope`, variable weights 400–700. Use for paragraphs, navigation, card descriptions, buttons, metadata that needs readability.
- **Mono / data labels:** `IBM Plex Mono`, weights 400–500, used sparingly for DOI labels, platform tags, dates, build/status labels, tiny trailer captions, and project metadata. Never use for long prose.

Fallback stacks: `Arial, Helvetica, sans-serif` or platform sans-serif; monospace `ui-monospace, SFMono-Regular, Consolas, monospace`.

Type rules:

- Hero name large and responsive: clamp to fit without horizontal scrolling; target 64–112 px desktop and 40–64 px phone depending on viewport.
- Body: 16–19 px with line-height 1.55–1.75. Main reading measure 60–75 characters.
- Headings follow a clear H1 → H2 → H3 structure, with no skipped levels for appearance.
- Avoid long uppercase passages. Mono labels should be short and nonessential.

- Underline inline links in long-form text or otherwise provide a strong non-color-only signal.

5.5 Layout and shape

- Content max width: 1200 px; reading column: 720 px; generous side margins at large screens.
- Desktop grid: 12 columns; tablet: 8; phone: 4. Use flexible CSS grid/flex, not fixed canvas dimensions.
- Spacing base: 4 px; main rhythm uses 8, 16, 24, 32, 48, 64, 96 px. Large section spacing must reduce gracefully on mobile.
- Cards: default radius 20–28 px, with room-specific variations kept within the shared component rules. Avoid over-rounded “pill UI” for everything.
- Borders: fine and quiet; use shadows sparingly and never as the sole way to distinguish components.
- Keep content width and line length comfortable on phones; prevent overflow from title strings, buttons, source-code labels, and long URLs.

5.6 Room visual languages

Home / foyer: atmospheric gradient/light field, bold name typography, one signature living visual motif. The motif can react subtly to pointer, scroll progress, or time, but content remains static and accessible without it.

Projects / workshop: Material UI principles and components for useful catalogue controls: clear filter chips, segmented filters, readable cards, status labels, clear actions. If using MUI React, create a custom theme overriding Roboto defaults, shapes, colors, focus states, density, and elevation to match this PRD. Do not use default MUI look unchanged. Project cards may use a different graphic composition per product, but share same anatomy.

Research / observatory: paper/reading surface, quiet blue accents, calm diagrams or timeline; optional restrained liquid-glass panel for a compact metadata/control layer only. Keep abstracts and methods as ordinary selectable text. Glass must not reduce text contrast or introduce unreadable blur.

Creative / studio: papercraft cues such as paper layers, deckled-like graphic edges (CSS/SVG, not text noise), book-spine compositions, and warm amber accents. Keep screen-reader reading order linear and avoid image-only text.

About / personal: warm, human, editorial, fewer effects, optional owner-approved portrait. Plum accent as a subtle thread.

Contact / invitation: calm, direct, coral accent. Make contact methods obvious, form fields labelled, and success/error state explicit.

Neural brutalism: use as a limited graphic treatment for signal paths, nodes, oversized crop, or sharp editorial composition. Avoid brutalist illegibility, flashing, aggressive motion, chaotic grids, or making every area visually loud.

5.7 Liquid-glass guardrails

Liquid glass is a decorative material for small surfaces such as the trailer controls, a floating navigation panel, or a project preview overlay. Use translucent fill, blur, hairline stroke, and subtle highlight. Provide an opaque fallback for unsupported browsers and `prefers-reduced-transparency` where available. Never put long body copy over moving or low-contrast glass. Keep click targets stable; don’ t make content appear only on hover.

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6. Motion and animation specification

6.1 Motion vocabulary

Use a small repeatable system:

- **Reveal:** opacity + 8–20 px vertical shift, 350–650 ms, ease-out.
- **Page/room transition:** crossfade or subtle shared-element/light sweep, 200–400 ms; do not delay navigation.
- **Card response:** 2–5 px lift or scale to 1.01–1.02, 160–240 ms; avoid large tilts that harm reading.
- **Living visual:** slow low-amplitude shape/light movement, 8–20 seconds per cycle, never required to understand content.
- **Trailer text:** cinematic cuts, short masks, line reveals, and paced typography; each informational frame remains long enough to read.
- **Scroll progress:** optional subtle section signal only. Do not hijack or snap the scroll position.

Use CSS transforms and opacity for most motion. Avoid continuous high-cost filters and canvas animation as background. Aim for smooth performance on current mid-range mobile devices. Avoid simultaneous competing animations.

6.2 Accessibility and user control

- Respect `prefers-reduced-motion: reduce`; replace movement with quick opacity changes or no animation.

- No flashing/flicker; never exceed WCAG flash thresholds.
- No autoplay audio. Trailer autoplay video only after explicit user action.
- Play, pause, mute, replay, skip/close controls visible and keyboard accessible.
- Ensure all meaning communicated in motion is also available as text.
- No forced device rotation anywhere in core site. Trailer may offer a non-blocking "Rotate for widescreen" hint on phones.

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7. Page-by-page requirements

7.1 Home (/)

Purpose: introduce Vijayarka, establish atmosphere and positioning, then provide immediate routes to the rest of the site.

Required content hierarchy:

1. Global header.
2. Hero with exact name: **Vijayarka Nanduri**.
3. Tagline and concise supporting line.
4. Primary CTA **Explore the work**; secondary CTA **Play the intro**.
5. One distinctive visual that suggests ideas evolving or connecting.
6. "Selected work" section with 3–5 editorially chosen projects; include one web/PWA, one Android, and one distinctive/experimental item when available.
7. Teaser links to Research and Creative.
8. Short About preview and Contact CTA.
9. Footer.

Draft positioning copy, owner approval required:

- Tagline option A: **"I build things that feel a little more alive."**
- Tagline option B: **"Research, software, and stories—with a human pulse."**
- Supporting line: **"I'm Vijayarka Nanduri—an independent researcher, builder, and writer exploring adaptive interfaces, human-centered AI, and creative technology."**

Do not put competing taglines together; choose one after brand review. Avoid several paragraphs in the hero.

7.2 Trailer / visual intro

Purpose: give a quick, optional, emotionally engaging introduction for people who prefer watching to reading.

Presentation: open in an accessible modal/full-screen stage, or dedicated `/intro/` route with a modal-like close/back path. Primary format 16:9; make a responsive stage that can display in portrait. In portrait, show an unobtrusive prompt such as **"Turn your phone sideways for the widescreen version"**, with **"Stay here"** and **"Skip intro"** actions. Never require rotation to continue or access underlying site.

Length: target 25–40 seconds. This is short enough to feel like a trailer, not a documentary. User can stop it immediately.

Storyboard (editable, no fabricated claims):

- 0–3 s: "Vijayarka Nanduri" — quiet light / signal motif.
- 3–8 s: "Researcher. Builder. Writer."
- 8–14 s: adaptive interfaces / Alive UI / MAWO visual.
- 14–20 s: personal software such as privacy-conscious tools (visuals from approved products only).
- 20–26 s: research, experiments, books/games/audio represented by original assets.
- 26–32 s: "Different ideas. One human point of view." (candidate line; owner to approve.)
- End card: **Explore projects / Read research / Get in touch / Replay**.

Do not state that every app is live; use only confirmed project states. The same story must exist in accessible captions/transcript and, ideally, a reduced-motion sequence.

Media delivery: use optimized WebM/MP4 with poster image, lazy-load before play, provide captions/subtitles if spoken narration is added, and include a lightweight text-only fallback. Do not generate/use a high-bandwidth video background as the sole hero. No sound unless visitor explicitly enables it.

7.3 Projects index (`/projects/`)

Purpose: a curated, own-app-store-like catalogue that lets visitors discover, understand, and launch projects.

Top section: clear page title ("Projects"), one-sentence explanation, category/filter controls, and featured project feature block. Avoid marketing claims that the catalogue includes every idea unless it does.

Filters:

- All

- Web & PWA
- Android
- Research prototypes / Experiments
- Creative coding / Games (if enough items)
- Status: Live, In development, Archived (optional filter)

Filters work by keyboard and touch, update result count accessibly, and have a clear reset. Provide URL query parameters or route state if reasonable, but ensure crawlable project pages independently exist.

Card anatomy: product image/screenshot (with useful alt text), name, category/platform tags, 1–2 sentence plain-language description, status, and one clear primary action plus secondary source link when available. Do not rely on hover for critical details.

Detail page anatomy:

- Project title + category/platform + honest status.
- One-sentence product pitch.
- Hero screenshot/demo with captions/alt.
- Problem or idea; what it does; why it exists.
- Vijayarka’ s role/contribution, phrased only from verified information.
- Key features, selected design/engineering decisions.
- Availability actions: try web/PWA, download APK, view repo, read research.
- Privacy/permissions/data note where documented and verified.
- Version/release date and last-checked field where useful.
- Related projects and next/previous browsing.

7.4 Research index and article pages

Purpose: make research findable and comprehensible, without inflating the status of preprints.

Index displays paper title, date, paper type (label the three Zenodo records **Preprint**), short plain-language summary, DOI link, and related project. Article page contains abstract/summary, question, approach, findings as reported, limitations, citation/DOI, PDF/source link, and related work. Clarify study size and status near claims.

Do not call the Zenodo records peer-reviewed or journal-published unless independently verified. For MAWO, identify the described 45-day self-study and explain that reported 71% → 96% completion is from that case study, not a general causal claim. For Self-Narrating Neural Networks, note the reported two-student proof of concept and limitations. For the

Draftbook paper, preserve the paper as research history, but do not imply the current product is unavailable; owner confirms it is working as of this brief.

7.5 Creative index (`/creative/`)

Show books, free guide, games, audio-fiction, and selected videos/writing as editorially curated objects. Use book covers, game imagery, and audio controls only if rights/owner approval allow. Link to current destinations. State work type and availability. Short descriptions should convey tone and premise, not overstate awards or audience size.

7.6 About (`/about/`)

Use a concise first-person bio, guiding interests, selected milestones, and a through-line from research to building and writing. A timeline can use dates only where public and approved. Avoid claiming degrees, employer status, or credentials without owner confirmation. Portrait is optional and requires approval. Allow a playful detail, but maintain professional readability.

7.7 Contact (`/contact/`)

Present a clear invitation and the types of contact welcome (research collaboration, product/design conversations, creative work—owner to confirm). Preferred implementation: a working mailto link plus professional social/contact profiles. If implementing a form, use a real backend or managed form provider; validate, show success/failure, prevent spam with an accessible method, and publish an accurate privacy notice. Do not add a fake form that silently does nothing. Do not show private email without approval.

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8. Project inventory and publishing rules

Treat this inventory as sourced draft content, not final approval. Preserve the user's corrections as controlling status information: **Draftbook is working again**; **Tofu Pet remains under development**; **Kosha is private/not yet announced and must not be published publicly until Vijayarka approves release**. Check all links and actual releases immediately before launch.

8.1 Primary / current public projects

MAWO
Category: Web/PWA and Android project/research case study. Pitch: a habit tracker whose progress is represented through evolving interface phases. Links: MAWO site, current launch app, GitHub, Alive UI paper. Status: confirm which URL is the canonical user-

facing live product; mark PWA and Android only if both current. Do not claim broadly "built on neuroscience" or general efficacy without appropriate evidence qualification.

Khata
Category: Android. Pitch: private, offline-first personal ledger for everyday spending, budgeting, and money habits. Links: product page, repository, latest release. README documents local data, CSV import/export, home-screen widgets, repeat-purchase nudges, and no bank linking; verify current release and permissions before restating technical/privacy promises.

Metabasis
Category: Android. Pitch: a step counter presented as a tactile instrument; movement becomes stages/progress. Links: repository and project site source, planned canonical page in repository metadata: `https://vijayarkananduri.github.io/metabasis/` (confirm it resolves before publishing), release instructions are documented in repo. README says data stays local and sensor behavior can vary; include limitations and say fallback counting may be approximate. Do not state universal accuracy.

Almara
Category: Android. Pitch: find phone files by time, source, type, or saved virtual folders without moving originals. Links: live project page, repository, Android releases, guide. README asserts no network permission, no analytics, and encrypted local index; verify the shipped release manifest and behavior before using absolute privacy promises.

ExamFlow
Category: Web/PWA. Pitch: private browser-based semester planner that turns syllabus topics and exam dates into a focused daily study plan. Link: repository. Verify and publish the actual live app URL from repository README/deployment settings before adding a "Try it" link. Repo contains `manifest.webmanifest` and `sw.js`, supporting PWA presentation; verify installability in deployed app before claiming PWA on the card.

Centrum OS
Category: Linux/CLI experiment/tool. Pitch: a personal terminal workflow tool for focus, agenda, project launching, and local notes. Link: repository. Status/version in README is v0.1 beta; label experimental/beta. Do not call it a consumer desktop operating system or imply a stable supported distribution without testing.

8.2 Portfolio legacy / other selected projects

Tenet (portfolio link): privacy-first emotion-tracking journal as described on current portfolio. Confirm live URL and implementation/privacy details.
DesignRx (portfolio link): AI-assisted website design auditing as described on current portfolio. Confirm live status and what AI actually does.
Draftbook (portfolio destination; research preprint): AI writing support for creative

writers. Status per Vijayarka: **working/live again**. Confirm live product URL and capture updated screenshot. Do not copy the old Zenodo down notice as current status; it describes an earlier version.
Echo (itch.io): sound-first browser horror game; verify playable status and accessibility alternatives.
Ek Indian Bacha (game): browser satire game; verify game link.
Logbook (project): local second-brain tool as described on the portfolio; verify live behavior and local-storage claims.
Writer's Assistant / Tailored Poems (portfolio Linktree): only feature if each destination is clearly identified, current, and working. Prefer direct app/repository/download links over a link hub when available.
Sylph (repository): weather/air quality app with animated conditions and outfit recommendations per README; include only if owner considers it a project worth featuring and current data APIs still work.

8.3 Work not yet public

Tofu Pet — user says under development. Do not present as launched, do not provide download CTA. Optional small "in development" feature can be published only with owner approval and the public repository link: Tofu Pet repo. Do not imply all listed README roadmap/behavior is shipped.

Kosha — user says it is planned to become public in a few days; supplied notes show the GitHub repository is currently private. Keep entirely out of public navigation, sitemap, search metadata, trailer, and public project pages until explicit public-release approval and verified URL/release. Prepare a draft project record only. Security wording must reflect the README's limitations: it says record payloads and attachments are encrypted locally, while the database itself is not SQLCipher-encrypted and the app is not claimed to defend a compromised/rooted device. Avoid absolute claims such as "unhackable" or "military-grade security."

8.4 Project prioritization

Homepage feature selection: choose three to five current works with visual distinction, user-approved availability, and working destinations. Suggested initial selection: MAWO, Almara, Metabasis, Khata, and Draftbook, with ExamFlow as a web/PWA feature in catalogue. This is a recommendation; owner should approve the homepage order. Tofu can be a clearly marked in-development teaser only with approval. Kosha enters the public project catalogue only on/after release.

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9. Search engine and sharing requirements

9.1 Technical SEO

- Server-rendered/static-rendered HTML or a build-time static site so content and metadata are crawlable without requiring browser-only JavaScript execution.
- Unique descriptive `<title>` and meta description for every route.
- Canonical URL using the final chosen domain; do not leave temporary Netlify/Pages URLs canonical if a custom domain is established.
- `sitemap.xml` listing only canonical, public, indexable routes. Never include private/unreleased Kosha pages or noindex staging routes.
- `robots.txt` that permits public pages and references sitemap; robots rules must not be used as a privacy mechanism.
- Open Graph and X/Twitter card metadata, with unique project/research preview images where possible.
- Descriptive internal links; no generic repeated "click here."
- One H1 per page, logical heading hierarchy, crawlable text, descriptive image alt, visible dates and content status.
- Working custom 404, HTTPS, mobile responsive, no broken canonical/redirect loops.
- Generate clean RSS/feed for writing/research if a maintained publishing path exists (optional).

9.2 Metadata examples

Adapt to deployed origin and final titles; avoid duplicate titles.

- Home title: `Vijayarka Nanduri — Research, Software & Creative Work`
- Home description: `Meet Vijayarka Nanduri, an independent researcher, builder, and writer exploring adaptive interfaces, human-centered AI, software, and creative work.`
- Projects title: `Projects by Vijayarka Nanduri — Apps, Web Tools & Experiments`
- Research title: `Research by Vijayarka Nanduri — HCI, Alive UI & AI`
- Creative title: `Creative Work by Vijayarka Nanduri — Books, Games & Audio`
- About title: `About Vijayarka Nanduri`
- Contact title: `Contact Vijayarka Nanduri`

9.3 Structured data

Use valid JSON-LD only when its properties match visible content:

- `Person` on About/home: `name`, `url`, approved `image`, `sameAs` to confirmed accounts, and accurate `knowsAbout`. Do not invent affiliations, credentials, awards, or location.
- `WebSite` for overall site.
- `CreativeWork` / `SoftwareApplication` only if the schema accurately represents the project and availability. Avoid fake ratings, prices, OS distribution claims, or ratings.
- `ScholarlyArticle` only for scholarly work page metadata if the record represents it accurately; include `creativeWorkStatus` /description as preprint where appropriate, do not imply publication venue not in evidence.

Structured data does not guarantee enhanced results. Do not add keyword stuffing or hidden text.

9.4 Brand entity consistency

Use the same approved name, avatar/logo, site URL, and social profile URLs across the site, GitHub profile, LinkedIn, ORCID, and Zenodo author references. Link out to canonical work records and back to personal site where appropriate. Keep research titles and DOI metadata exact.

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10. Accessibility, responsiveness, and performance

Accessibility target

Target **WCAG 2.2 AA** for the public site. Minimum requirements:

- Text contrast at least 4.5:1 for normal text, 3:1 for large text; interactive component boundaries/focus indicators meet applicable non-text contrast.
- All actions keyboard-operable and visible focus.
- Semantic landmarks, labelled form inputs, correct heading hierarchy, meaningful alt text, captions/transcripts for meaningful video/audio.
- No information conveyed only by color, movement, hover, or sound.
- Respect reduced motion; no autoplay audio; pause control for long-running movement.
- Touch targets ideally at least 44 × 44 CSS px, especially navigation/filter/video controls.
- Accessible modal focus management and escape-to-close.
- Text zoom/reflow usable to 200% and at narrow width without horizontal scrolling except intentional media.

Responsive behaviors

Test at minimum: 360×800, 390×844, 768×1024, 1024×768, 1440×900, and wide desktop 1920×1080. Test portrait and landscape phone, tablet, desktop. No fixed hero height that clips content. On mobile, preserve narrative hierarchy; convert grids to stacked cards, place controls within reach, use project preview imagery at useful crop, and never require hover.

Performance targets

- Target Core Web Vitals “good” thresholds on representative mobile tests after assets are optimized.
- Initial content appears quickly on ordinary mobile network; avoid shipping all screenshots/video in first load.
- Use responsive image sizes, modern formats (AVIF/WebP with fallback), explicit dimensions, lazy-load below-fold media, and preload only truly critical hero assets/fonts.
- Trailer video is downloaded only after user chooses Play (or near the point it is opened), with lightweight poster.
- Avoid unnecessary JS, blocking third-party trackers, huge animation libraries, and web font layout shifts.
- Keep essential copy and routes available if JavaScript fails where feasible.

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11. Recommended implementation approach

This PRD is implementation-flexible, but a reliable static-first build is preferred:

- React + TypeScript with Vite or equivalent static generation; choose one, do not mix multiple stacks.
- Use MUI only where it adds value (project filters, dialogs, forms), with a fully customized theme matching these design tokens. If MUI adds excess bundle/visual weight, implement simple accessible components in the same design system rather than using defaults.
- CSS variables for global design tokens, modular CSS or a coherent styling solution, and one reusable component library for Header, Footer, Button, ProjectCard, StatusBadge, FilterBar, MediaFrame, ResearchCard, and Modal.
- Static/content-first data files for portfolio projects and research pages so status and links can be updated centrally. Content types should include title, slug, one-line summary, category, platform, status, screenshot/alt, dates, primary CTA label/URL, source URL, related records, and last-verified date.

- No database, authentication, or backend is required for the basic personal site. Contact form is optional; use a real provider if implemented.
- Deployable as a static site on Netlify or another host with route fallback/redirect rules, HTTPS, sitemap, and custom domain configuration.
- Include a README with local development, build, deploy, content-update, environment variable, and asset/license instructions.
- No secrets, API keys, personal data, private repository contents, or private Kosha assets in source or public bundle.

Asset policy

Use only owner-approved images, screenshots, book covers, logos, and portraits. Prefer actual product screenshots over generic AI-generated product UI. If custom abstract art is generated, keep it as decorative atmosphere, not as a fake screenshot or false evidence of a feature. Include alt text or mark purely decorative images with empty alt. Confirm rights/licenses and attribution for all fonts, sound, and external media.

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12. Content and factual QA rules

Before deployment, Vijayarka or an authorized reviewer must approve:

- the final tagline, short bio, role labels, and about-page biography;
- which projects are featured and their live/development/archived status;
- links, platform labels, download destinations, repository ownership, and product images;
- claims about privacy, offline use, data storage, sensors, encryption, research evidence, and performance;
- external email/contact details and social profiles;
- Kosha public announcement and its safe, accurate security description;
- Tofu development status language and whether it may be publicly teased;
- dates, timeline entries, publication type, DOI, research limitations, book listing, and creative links.

Do not carry forward stale old-site counters such as total papers, downloads, reads, years active, or project count without a date, a transparent counting method, and a current source. Avoid declaring “published papers” if the correct status is preprint. Draftbook’s current availability is provided by the owner as working; verify the exact production URL

during content freeze. Never use the older research-page outage notice as a current site status.

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13. Analytics and privacy

No tracking is required for launch. If analytics are introduced, use a privacy-conscious configuration, document what it collects, honor consent/law applicable to target visitors, and do not send personal project/private-use data. Project sites may have their own privacy behaviors; do not imply that the personal brand site has the same data model as those products. A portfolio page describing an app’ s privacy must distinguish app behavior from website behavior.

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14. Definition of done / acceptance criteria

The site is ready to publish only when all critical criteria pass:

Brand and usability

- The home hero displays **Vijayarka Nanduri** and one owner-approved tagline without awkward wrapping at all target viewports.
- Visitors can understand who he is and find Projects, Research, Creative, About, and Contact in one clear navigation system.
- Play intro is optional; normal navigation and content remain immediately available.
- No dead buttons, fake forms, placeholder lorem ipsum, broken links, or unlabeled controls.

Visual quality

- One shared design system is visible throughout; room differences are accents/materials, not random redesigns.
- Material components are themed; no unmodified default MUI style.
- Glass effects never cover long reading text; paper elements remain legible; brutalist graphic language does not impair comprehension.
- Animations enhance rather than block understanding; site remains attractive with motion disabled.
- Real project imagery or explicitly decorative visuals are used; no misleading mock product screenshots.

Responsive and accessible

- Layout tested across listed screen sizes; no unintended horizontal overflow or clipped CTA.
- Keyboard-only user can navigate, operate filters, use/close trailer, and access contact.
- Reduced motion works; trailer sound is off until user opts in; caption/transcript available if speech exists.
- Contrast and focus states checked against WCAG 2.2 AA target.
- Mobile rotation is suggested only for the 16:9 trailer and can be skipped.

Content and privacy

- Draftbook is shown as working/live only after the exact current URL is tested.
- Tofu is not represented as launched; any preview is owner-approved and labelled in development.
- Kosha has no public page, link, asset, metadata, sitemap entry, or preview until public-release approval.
- Claims about MAWO/research, privacy, offline support, encryption, or app accuracy are accurately qualified and tied to evidence.
- No private repository material, unpublished source, credentials, or unapproved contact information appears in the public bundle.

SEO and deployment

- Every public route has unique title, description, canonical, Open Graph image, and correct indexability.
- Sitemap contains only valid public canonical URLs; robots and redirects work; 404 is useful.
- Page content is available as crawlable HTML, not hidden behind an animation or client-only interaction.
- Production build and host deployment succeed; HTTPS works; images/fonts/video load from correct paths; share previews are tested.
- No console errors, missing assets, accessibility-blocking runtime errors, or unexpected third-party requests.

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15. Recommended build sequence

1. Confirm name/tagline/owner-approved copy and project statuses. Choose canonical domain and hosting.
2. Build shared foundations: semantic layout, tokens, fonts, navigation, responsive grid, buttons, focus states, footer.
3. Build homepage static composition first for desktop and mobile; review the identity before adding effects.
4. Implement project catalogue and reusable detail template. Add only verified project data and actions.
5. Implement Research and Creative index/detail templates with correct status and citations/links.
6. Add About and Contact; connect only approved contact paths.
7. Implement optional trailer and mobile landscape invitation; test sound, captions/transcript, skip, keyboard, reduced motion, and portrait fallback.
8. Add SEO metadata, sitemap, structured data, sharing images, redirects, and branded 404.
9. Test the complete public site on real phone and laptop; validate links and claims; perform owner content review.
10. Deploy to staging, inspect share previews and indexability, then publish. Add Kosha only as a separate approved post-release content update.

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16. Builder instruction (copy/paste prompt)

Build the complete website specified in this PRD. Treat the PRD as the source of truth. First create the shared responsive design system and all required routes/components; then implement the content and behaviors. The result must be a complete, accessible, deployable website, not a static mockup. Use only confirmed content and provided links; do not invent projects, claims, credentials, publication status, metrics, or contact details. Treat Draftbook as working per owner confirmation but verify the production URL; label Tofu as under development and do not present it as downloadable; omit Kosha completely from public pages until explicit approval for its public release. Preserve normal navigation even when the optional trailer is available. On mobile, offer—but never require—rotation for the 16:9 intro. Follow the exact color and typography tokens, responsive, motion, accessibility, SEO, privacy, and acceptance criteria in this document. Add a README with build/deploy instructions and a concise list of any unresolved owner confirmations. Before calling the build finished, test the site at the specified viewports and check links, keyboard use, reduced motion, production build, SEO metadata, and no console errors.

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17. Sources and project links

These primary sources were consulted for the draft inventory and factual notes. Current status and deployment destinations must still be checked during content freeze.

- Current portfolio
- GitHub profile and repositories
- MAWO repository
- MAWO live landing page
- Khata repository
- Khata product page
- Metabasis repository
- Almara repository
- Almara public page
- ExamFlow repository
- Centrum OS repository
- Tofu Pet repository
- Sylph repository
- Alive UI / MAWO Zenodo preprint
- Self-Narrating Neural Networks Zenodo preprint
- Emotion-Aware AI Writing Tools Zenodo preprint
- Draftbook product link from portfolio
- LinkedIn profile
- ORCID profile